This booklet is part of a continuing patient education series on the thyroid gland and thyroid disease. Additional booklets available in this series are:

Getting to Know Your Medication
How the Thyroid Gland Can Affect Your Weight
Hypothyroidism and Your Health
The Postpartum Period and Your Thyroid
Thyroid Disease and Aging
The Thyroid Gland and Your Health
Thyroid Disease and Diabetes
Thyroid Disease in Women
Understanding Goiter
Understanding Graves' Disease
Understanding Thyroid Cancer
Understanding Thyroid Nodules
Understanding TSH Testing
Understanding Your Child's Thyroid
What Is Hashimoto's Disease?

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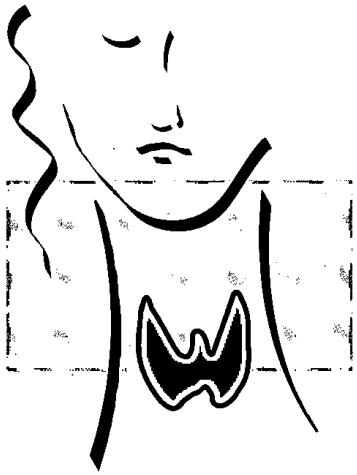
Hyperthyroidism and Your Health



Hyperthyroidism— What You Need to Know

Hyperthyroidism is a condition in which your thyroid gland makes too much thyroid hormone. Since thyroid hormone affects nearly every organ, tissue, and cell in your body, hyperthyroidism can interfere with most of your body's functions. Fortunately, there are a variety of reliable treatments available for hyperthyroidism, and with the proper care, you can lead a normal, healthy life.

The following pages contain basic information about the thyroid gland, its role in your body, and thyroid disorders. In addition, this booklet will discuss hyperthyroidism.

Remember, no single booklet can answer all of your questions or replace speaking with doctors, nurses, and other members of the healthcare team. If you have any questions or concerns about the thyroid or thyroid disorders, please check with your doctor.

Notes:

For more information on the thyroid gland, please contact:

The Thyroid Foundation of America
Ruth Sleeper Hall, RSL 350
40 Parkman Street
Boston, MA 02114-2698
(send self-addressed, stamped envelope)

The Thyroid Society for Education & Research
7515 S. Main Street, Suite 545
Houston, TX 77030
1-800-THYROID

National Graves' Disease Foundation
2 Tsitsi Court
Brevard, NC 28712

Or visit our web site at:
www.glandcentral.com

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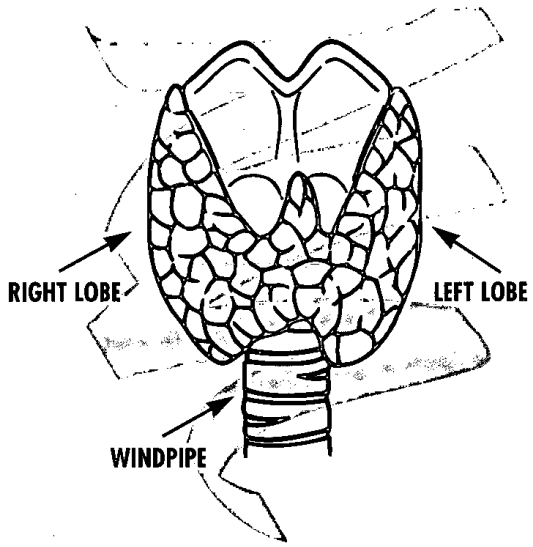
Your Thyroid— The Basics

Your thyroid is a gland located at the base of your neck, just below your Adam's apple. It is shaped like a butterfly – each "wing," or **lobe**, of your thyroid lies on either side of your windpipe.



The purpose of your thyroid gland is to make, store, and release **thyroid hormone** into your blood. Thyroid hormone affects essentially every cell in your body, and helps control your body's functions. If you have too little thyroid hormone in your blood, your body slows down. This is a condition called **hypothyroidism**. If you have too much thyroid hormone in your blood, your body speeds up. This is a condition called **hyperthyroidism**.

A Final Note

While finding out that you have hyperthyroidism may seem overwhelming, there are very effective treatments available to you. Additionally, finding out as much as you can about your condition can help you communicate with your doctor and address any concerns you may have in deciding which treatment may best fit your particular condition. With your doctor's expertise and your cooperation, you can overcome your thyroid condition and enjoy a normal, healthy life.

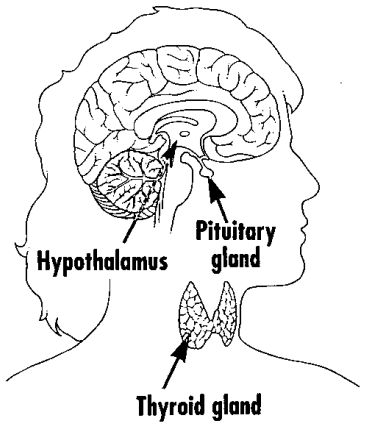
Maintaining Your Health

Seeing your doctor for periodic examinations and TSH tests is the best way to keep your thyroid hormone level normal. Over your lifetime, your body's need for thyroid hormone may change, and the safest way to prevent the damaging effects of abnormal amounts of thyroid hormone is to have your doctor measure your thyroid function regularly so that he or she can adjust your dose to meet your current needs.

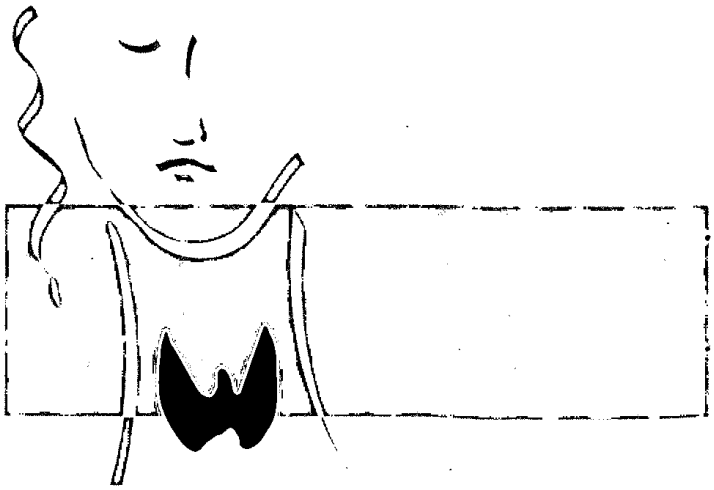
The amount of thyroid hormone made by your thyroid gland is adjusted by another gland called the **pituitary**, as well as a part of your brain called the **hypothalamus**.

- The pituitary gland makes hormones that control many other glands in your body
- The hypothalamus helps the pituitary control many of your glands, as well as helping to control other bodily functions, such as thirst, hunger, sleep, and body temperature

The thyroid gland, pituitary gland, and hypothalamus all work together to control the amount of thyroid hormone in your body. With the pituitary controlling most of the action, these structures work in a way that is similar to the way a thermostat controls the temperature in a room.



For instance, just as the thermometer in the thermostat senses the temperature of a room, your **pituitary** constantly senses the amount of thyroid hormone in your blood. If there is not enough thyroid hormone, your pituitary senses the need to "turn on the heat." It does this by releasing more **Thyroid-Stimulating Hormone (TSH)**, which signals your thyroid to make more thyroid hormone. Your thyroid gland then makes and releases the hormone directly into your bloodstream and restores the amount of thyroid hormone in your blood to normal.



Don't ever stop taking your thyroid hormone replacement therapy without talking to your doctor. If you stop taking it, you may continue to feel fine for a while. Nevertheless, your hypothyroid symptoms will likely return.

Doing Your Part

If you do become hypothyroid and need thyroid hormone replacement therapy, it is very important that whenever you get your levothyroxine sodium prescription refilled, you check to be sure that you are receiving what your doctor ordered. It is a good idea to take the same brand of thyroid hormone that your doctor started you on, because even the same dose of another brand may not deliver the identical amount of levothyroxine sodium. Do not switch brands without checking with your doctor.

If you begin to have symptoms of fatigue, weight gain, or any other signs of hypothyroidism, tell your doctor. You may need a higher dose.

Similarly, if you notice that you have symptoms of hyperthyroidism, there is a chance that your dose may be too high. Make an appointment to see your doctor right away if this occurs.

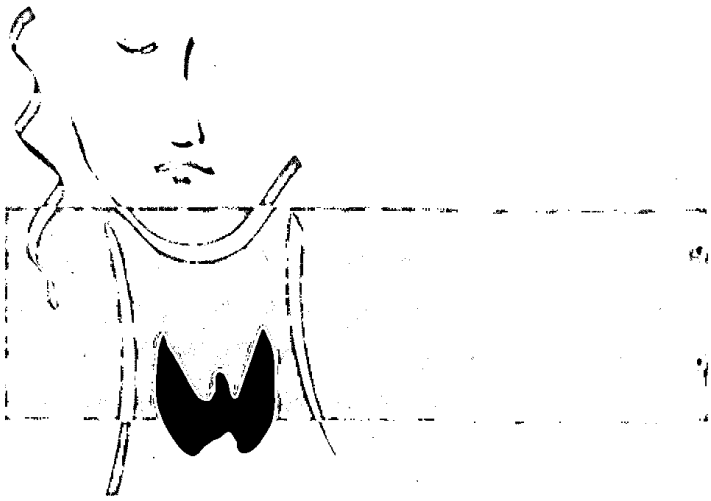
It is extremely important to get exactly the right amount of thyroid hormone. Too much or too little may place your body at risk for developing other conditions that can affect your health.

Your pituitary then senses that there is just the right amount of thyroid hormone in your body. With your thyroid hormone levels now restored to normal, your pituitary slows its production of TSH back down to normal.

Thyroid Disorders— A Brief Look at What Can Go Wrong

Hypothyroidism

If your thyroid gland produces too little thyroid hormone, you may have a condition called **hypothyroidism**. If you are hypothyroid, you also have a high level of TSH (which is your body's signal to "turn on the heat," or to step up the production and release of thyroid hormone). Having too little thyroid hormone causes your body to slow down. As a result, you may feel tired, have a slow heartbeat, or feel depressed, besides having other symptoms.



Surgery

- Your doctor may recommend that part of your thyroid gland be removed by surgery. This procedure is known as a **thyroidectomy**. Surgery is the least frequently used type of treatment, but may be chosen if you:
  - are pregnant and don't respond or are allergic to antithyroid drugs
  - do not wish to have radioactive iodine therapy
  - also have a thyroid nodule

Very often after being treated for hyperthyroidism, you will become hypothyroid and need to take a **synthetic** (man-made) thyroid hormone called **levothyroxine sodium** (the treatment for hypothyroidism), usually for the rest of your life. This is because your remaining active thyroid cells cannot make enough thyroid hormone.

Fortunately, thyroid hormone replacement is a simple therapy. By faithfully taking your medication **every day** and reporting any symptoms you have to your doctor promptly, you can keep your thyroid disorder under control.

Treating Your Hyperthyroidism

There are several ways to treat hyperthyroidism. Your doctor will explain the options, and may recommend a particular course of treatment that seems best suited to your needs and lifestyle. The following are some of the options:

Antithyroid Drugs

- These are medications that you take in tablet form. They slow down your thyroid's production of thyroid hormone. Often, however, your hyperthyroidism will return after the drugs are stopped, and you will need further treatment

Radioactive Iodine

- Although it may sound alarming, radioactive iodine is a safe and effective treatment for hyperthyroidism. The substance enters your thyroid gland and permanently destroys thyroid cells. That way, those cells can no longer make thyroid hormone. If you undergo this course of treatment, you may become hypothyroid. In that case, you will need thyroid hormone replacement therapy

Mild Thyroid Failure

A more mild form of hypothyroidism is **mild thyroid failure**. In this case, your thyroid hormone levels are normal, but you have a high TSH level. This may mean that TSH is making your thyroid gland work extra hard just to maintain normal levels of hormone in your blood. This condition often leads to hypothyroidism. People with mild thyroid failure generally don't show any obvious symptoms, which makes it very difficult for doctors to identify without performing a TSH test.

Hyperthyroidism

If your thyroid gland produces too much thyroid hormone, you may have a condition known as **hyperthyroidism**. If you are hyperthyroid, you also have a low level of TSH in your blood because your pituitary has decreased its TSH signals to your thyroid gland. Having too much thyroid hormone causes your body to speed up, resulting in nervousness, fast heartbeat, and unexplained weight loss, among other symptoms.

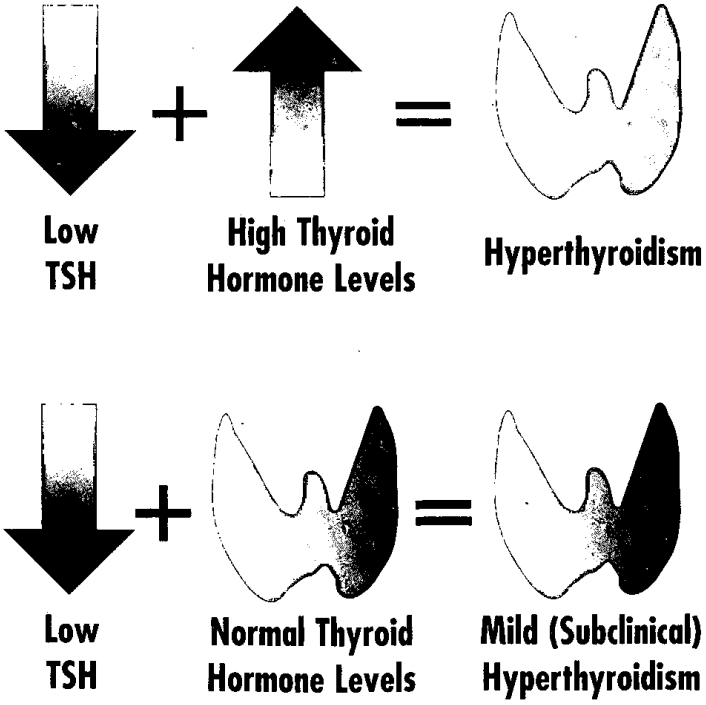
- Other tests that your doctor may order are:

  –**Thyroid antibody test**. This test will help your doctor find out if your immune system is affecting your thyroid function (as in Graves' disease)

  – **Radioactive iodine uptake test**. Radioactive iodine is a substance that you swallow in capsule form. Between 6 to 24 hours later, your doctor uses a special piece of equipment to measure how much of the radioactive iodine your thyroid absorbed. You may also have an image taken of your thyroid called a **thyroid scan**. Your doctor can use this information as a guide toward a correct diagnosis

Your Laboratory Results

- Your doctor will likely order blood tests to find out how your thyroid gland is functioning. The best single test is the **TSH test**. If your TSH level is lower than normal, it is likely that you have hyperthyroidism. Your doctor will probably measure your **thyroid hormone levels** as well, in order to determine the severity of your hyperthyroidism. If your TSH level is low and your thyroid hormone levels are normal, you may have a mild form of hyperthyroidism, called mild (subclinical) hyperthyroidism



Mild (Subclinical) Hyperthyroidism

If your thyroid hormone levels are normal, but you have a low TSH level, you may have a condition known as **mild (subclinical) hyperthyroidism**. This often progresses to hyperthyroidism at a later point, and should be monitored carefully by your doctor.

Hyperthyroidism— A Closer Look

Hyperthyroidism is a fairly common thyroid disorder, and results from too much thyroid hormone being made and released into your bloodstream. This can be caused by a number of things, including:

- ***Graves' Disease***

Graves' disease is a condition in which the **immune system** causes the thyroid gland to produce too much thyroid hormone. On average, women are five times more likely than men to have this disease

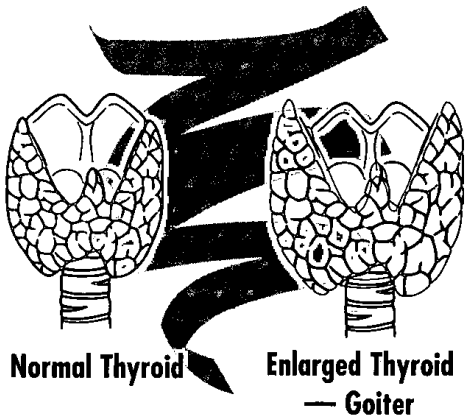
- ***Thyroid Nodules***

Hyperthyroidism caused by nodules simply means that there are **lumps** in the thyroid gland that are overproducing thyroid hormone

- ***Overreplacement of Thyroid Hormone***

A patient receiving **thyroid hormone replacement therapy** for *hypo*thyroidism and given too high a dose (or too much thyroid hormone) can become *hyper*thyroid

- In particular, there is one very reliable sign that your doctor will look for: enlargement of the thyroid, which is called a **goiter**. A goiter is a common physical sign of thyroid disorders. Your doctor will examine your neck, pressing on, or **palpating**, the area where your thyroid is located in order to determine if a goiter is present



- Your doctor will also examine your thyroid gland to determine if there are thyroid nodules present. Thyroid nodules are lumps in the thyroid gland that can result in too much thyroid hormone in your bloodstream. Sometimes, but very rarely, nodules can be cancerous

- In addition, your weight, blood pressure, and pulse will be measured

- Your doctor may examine your eyes, skin, heart, and nervous system, as well. All of these systems are affected by hyperthyroidism

Your Likelihood of Developing a Disorder

Some individuals are more likely than others to be hyperthyroid. You are at an increased risk if you:

- ✔ are a woman, 30 years or older
- ✔ are pregnant or have given birth in the last 6 months*
- ✔ have other **autoimmune conditions**, such as:
  - —Addison's disease
  - —Type I diabetes
  - —Pernicious anemia
  - —Rheumatoid arthritis
  - —Systemic lupus erythematosus (SLE or lupus)

*__Note:__ If you've had a baby in the past 6 months, you may be at a higher risk for developing hyperthyroidism due to autoimmune thyroid disease.

Your Physical Condition

- Your doctor will probably give you a physical exam to look for the signs and symptoms of hyperthyroidism

- ***Postpartum Thyroiditis***

This is a condition that occurs after giving birth and may include both hyperthyroidism and hypothyroidism. Postpartum thyroiditis can be a temporary condition or it can develop into long-term hypothyroidism

Having too much thyroid hormone in your blood causes your whole body to speed up. Your heart beats faster, you may feel nervous or jittery, you may be described by others as "irritable," and you may lose weight unexpectedly. Over the long term, untreated hyperthyroidism can cause very serious health problems. Therefore, it is very important that hyperthyroidism be identified and treated as soon as possible.

Identifying Hyperthyroidism

Not everyone with hyperthyroidism will have the same signs and symptoms. In fact, some people may have only a few, and some people may not have any. Moreover, some symptoms are often mistaken for other conditions. Keeping that in mind, the following is a list of ***possible*** signs and symptoms that you may experience if you have hyperthyroidism.

- Nervousness and irritability
- Sleep disturbances (for example, insomnia)
- Fast, irregular heartbeat
- Shortness of breath, especially with exercise
- Heat intolerance and increased sweating
- Shaky hands (tremor)
- Weight loss
- Changes in appetite (usually increased appetite)
- More frequent bowel movements
- Tiredness and muscle weakness
- Thyroid enlargement (goiter)
- Menstrual problems (decreased menstrual flow)
- Impaired fertility
- Eye irritation, double vision, bulging eyes

Making a Diagnosis

If your doctor suspects you have hyperthyroidism, he or she will probably investigate the following things.

Your Personal and Family Health History

Thyroid disorders can often be **hereditary**. If you have a family member, like your mother or father, an aunt or uncle, or your brother or sister, who has a thyroid disorder (especially if it is autoimmune), your chance of developing one may be higher.